



# Byte2Bit SEG-Y Quick User Guide

Simple Windows workflow for compressing SEG-Y files, restoring originals, and extracting random-access traces.

## 1. Open PowerShell in the Tool Folder

This guide uses the Windows executable `atlasSGY.exe`. Put `atlasSGY.exe`, your SEG-Y files, and `verify_random_trace.py` in a working folder, then open PowerShell in that folder.

```
cd "C:\path\to\your\folder"  
dir .\atlasSGY.exe
```

Every command below starts with `.\atlasSGY.exe` because PowerShell runs executables from the current folder using `.\`.

## 2. Check the License

Before compressing data, confirm that the license is visible and valid.

```
.\atlasSGY.exe license status
```

If the command reports that the license is active, the software is ready to use. If it reports a license problem, check that the license file or license configuration supplied with the demo is in the expected location.

## 3. Compress Like a Single Archive

Use normal compression when you want one compressed file that behaves like a zip-style storage container. This is useful for moving, sharing, or storing a full SEG-Y file in a smaller Byte2Bit-compressed form.

```
.\atlasSGY.exe compress input.sgy compressed.sgy
```

Replace `input.sgy` with your original SEG-Y file name. The output file `compressed.sgy` contains the compressed data.

## 4. Decompress Back to SEG-Y

Decompression restores the compressed file back to a standard SEG-Y file.

```
.\atlasSGY.exe decompress compressed.sgy restored.sgy
```

The restored file can be used by normal SEG-Y tools and workflows. For a full-file roundtrip test, compare `input.sgy` and `restored.sgy` with your existing validation process.

## 5. Compress for Random Access

Use `--randomAccess` when the compressed data should be optimized for trace-level access and cloud storage.

```
.\atlasSGY.exe compress input.sgy compressed.b2b --randomAccess
```

This creates a Byte2Bit file designed so software can access selected traces without decompressing the whole SEG-Y file first. For cloud workflows, this can reduce storage cost and egress cost because only the needed part of the compressed data has to be read.

Choose this mode for interactive interpretation, quality checks, cloud object storage, and workflows that repeatedly read only small parts of a large seismic file.

## 6. Restore a Random-Access File

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A random-access .b2b file can still be decompressed back to the original SEG-Y file.

```
.\atlasSGY.exe decompress compressed.b2b restored.sgy
```

Use this when a downstream tool needs the complete SEG-Y file rather than a selected trace.

## 7. Extract One Trace

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Random-access compression lets you decompress only the trace you need. This avoids restoring the whole seismic file when you only want one trace for inspection, testing, or transfer.

```
.\atlasSGY.exe extractTrace compressed.b2b trace-1000.sgy --trace 1000
```

This writes trace number 1000 to `trace-1000.sgy`. Change the trace number and output file name for other traces.

## 8. Verify an Extracted Trace

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The included `verify_random_trace.py` script compares a one-trace SEG-Y file against the same trace in the original SEG-Y file. It prints the first samples and reports whether the extracted trace matches byte-for-byte.

```
python .\verify_random_trace.py input.sgy trace-1000.sgy 1000
```

A successful result prints `PASS`. Use the same trace number that was passed to `extractTrace`.

## 9. Quick Test Workflow

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Run this sequence to try the main features with one SEG-Y file:

```
.\atlasSGY.exe license status
.\atlasSGY.exe compress input.sgy compressed.sgy
.\atlasSGY.exe decompress compressed.sgy restored.sgy
.\atlasSGY.exe compress input.sgy compressed.b2b --randomAccess
.\atlasSGY.exe extractTrace compressed.b2b trace-1000.sgy --trace 1000
python .\verify_random_trace.py input.sgy trace-1000.sgy 1000
```

## 10. Troubleshooting

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If PowerShell says the command is not recognized, make sure you are in the folder that contains `atlasSGY.exe` and include the leading `.\`.

- If license status fails, check the license file or license setup supplied with the executable.
- If compression fails, confirm the input file exists and is a valid SEG-Y file.
- If trace extraction fails, confirm the file was created with `--randomAccess` and that the requested trace number exists.
- If verification fails, make sure the original file, extracted trace file, and trace number all match the same test.